# Two brackets, one quadratic

$$x^2 + 5x + 6$$

$$(x + \square)(x + \square)$$

The coefficient of $x^2$ is 1. What belongs in the two spaces?

**? Which operation lets you check a proposed factorisation?**

# Multiply every term inside the bracket

$$x(x+3) = x \cdot x + x \cdot 3$$

$$= x^2 + 3x$$

The outside x multiplies both inside terms.

**? Which expansion equals 2(x+3)? A: 2x+3 B: 2x+6 C: x+6 D: 6x**

# Four products fill the grid

$$(x+2)(x+3)$$

| × | x | +3 |
|---|---|---|
| x | $x^2$ | 3x |
| +2 | 2x | 6 |

Each row term multiplies each column term.

Four products. Keep every sign.

# The middle terms describe the same power

$$x^2 + 3x + 2x + 6$$

$$= x^2 + 5x + 6$$

3x+2x=5x. The x² term and the constant stay separate.

# Reverse the multiplication

$$x^2 + 5x + 6 = (x + m)(x + n)$$

$$m + n = 5 \qquad mn = 6$$

The same two numbers must satisfy both conditions.

# Find a pair with product 6 and sum 5

| Pair | Product | Sum |
|---|---|---|
| 1 and 6 | 6 | 7 |
| 2 and 3 | 6 | 5 |

2 and 3 pass both checks. 1 and 6 pass only the product check.

# Build the factors and check the result

$$x^2 + 5x + 6 = (x + 2)(x + 3)$$

$$(x + 2)(x + 3) = x^2 + 3x + 2x + 6$$

The expansion returns x²+5x+6, so the factorisation is correct.

**? In (x+2)(x+3), which operation produces the constant 6?**

# Why the sum and product test works

$$(x+m)(x+n) = x^2 + nx + mx + mn$$

$$= x^2 + (m+n)x + mn$$

$$m + n = b \qquad\qquad mn = c$$

**? For $x^2+bx+c$, which target is the sum, and which target is the product?**

# Decide the signs before searching

| Constant c | Middle coefficient b | Signs of m and n |
|---|---|---|
| Positive | Positive | Both positive |
| Positive | Negative | Both negative |
| Negative | Positive or negative | Opposite signs |

With opposite signs, the number with larger absolute value decides the sign of the sum.

**? If c>0 and b<0, which signs work? A: +,+ B: −,− C: +,− D: either number is 0**

# Search systematically for a new pair

$$x^2 + 7x + 12$$

$m + n = 7$ $\qquad$ $mn = 12$

Both numbers are positive. List factor pairs of 12 in order.

# One factor pair passes both checks

| Pair | Product | Sum |
|---|---|---|
| 1 and 12 | 12 | 13 |
| 2 and 6 | 12 | 8 |
| 3 and 4 | 12 | 7 |

Stop when the two conditions match. Reversing a pair gives the same factors.

# Complete the second factorisation

$$x^2 + 7x + 12 = (x + 3)(x + 4)$$

$$x^2 + 4x + 3x + 12 = x^2 + 7x + 12$$

Try the next pair yourself: product 15 and sum 8.

**? Which numbers fit? A: 1 and 15 B: 2 and 6 C: 3 and 5 D: −3 and −5**

# Positive product, negative sum

$$x^2 - 5x + 6$$

$m + n = -5$ $\qquad$ $mn = 6$

A positive product gives matching signs. The negative sum makes both signs negative.

# Keep the negative signs in the search

| Pair | Product | Sum |
|---|---|---|
| −1 and −6 | 6 | −7 |
| −2 and −3 | 6 | −5 |

$$(-2)(-3) = 6 \qquad -2 + (-3) = -5$$

# Two negative constants give these brackets

$$x^2 - 5x + 6 = (x - 2)(x - 3)$$

$$(x - 2)(x - 3) = x^2 - 3x - 2x + 6$$

The two middle terms combine to −5x.

**? What creates the positive constant 6?**

# A correct product can hide a wrong sum

$$x^2 - 5x + 6 \stackrel{?}{=} (x - 1)(x - 6)$$

*A.* Both checks pass

*B.* The product is wrong

*C.* The sum is $-7$

*D.* The sum is $+7$

**? Which diagnosis is correct?**

# A negative product needs opposite signs

$$x^2 + x - 6$$

| Pair | Product | Sum |
|---|---|---|
| −1 and 6 | −6 | 5 |
| 1 and −6 | −6 | −5 |
| −2 and 3 | −6 | 1 |
| 2 and −3 | −6 | −1 |

**? The coefficient of x is 1. Which pair has sum 1?**

# Collect the positive and negative middle terms

$$x^2 + x - 6 = (x + 3)(x - 2)$$

$$(x + 3)(x - 2) = x^2 - 2x + 3x - 6$$

−2x+3x=+x. Keep the constant product −6.

CHOOSE ONE ANSWER

# Try another opposite-sign pair

$$x^2 + 2x - 8$$

$A.\quad (x+4)(x-2)$

$B.\quad (x-4)(x+2)$

$C.\quad (x+4)(x+2)$

$D.\quad (x+8)(x-1)$

**? Which factorisation has product −8 and middle coefficient +2?**

# Change the sum from +1 to −1

$$x^2 - x - 6$$

| Pair | Product | Sum |
|---|---|---|
| 3 and −2 | −6 | +1 |
| 2 and −3 | −6 | −1 |

**? Which number needs the larger absolute value when the sum is negative?**

# The middle coefficient fixes the order of signs

$$x^2 - x - 6 = (x + 2)(x - 3)$$

$$(x + 2)(x - 3) = x^2 - 3x + 2x - 6$$

The middle terms now combine to −x.

# Moving a minus sign changes the polynomial

$$(x+3)(x-2) = x^2 + x - 6$$

$$(x-3)(x+2) = x^2 - x - 6$$

The constant stays −6. The middle coefficient changes sign.

**? Which check detects this difference? A: leading coefficient B: product only C: sum D: constant only**

CHOOSE ONE ANSWER

# Use both targets without a factor table

$$x^2 - 2x - 8$$

$A.\quad (x-4)(x-2)$

$B.\quad (x+4)(x-2)$

$C.\quad (x-4)(x+2)$

$D.\quad (x-8)(x+1)$

**? Which factorisation has product −8 and sum −2?**

# A factor pair can use the same number twice

$$x^2 + 6x + 9$$

$$3 + 3 = 6 \qquad 3 \times 3 = 9$$

$$x^2 + 6x + 9 = (x + 3)(x + 3) = (x + 3)^2$$

# Squaring a bracket includes a middle term

$$(x+3)^2 = x^2 + 3x + 3x + 9$$

$$= x^2 + 6x + 9$$

When $(x+3)^2=0$, both equal factors give the same root $x=-3$.

**? How many distinct real roots does $(x+3)^2=0$ have? A: 0 B: 1 C: 2 D: 3**

# A missing middle term means b=0

$$x^2 - 9 = x^2 + 0x - 9$$

$$m + n = 0 \qquad mn = -9$$

Equal absolute values and opposite signs give a zero sum.

# The cross terms cancel

$$x^2 - 9 = (x - 3)(x + 3)$$

$$(x - 3)(x + 3) = x^2 + 3x - 3x - 9$$

The same sum-and-product rule explains the difference of two squares.

**? Which factors give x²−16? A: (x−4)² B: (x+4)² C: (x−8)(x+2) D: (x−4)(x+4)**

# A zero constant leaves a common factor

$$x^2 + 5x = x^2 + 5x + 0$$

$$x^2 + 5x = x \cdot x + x \cdot 5$$

Both terms contain x. Take that common factor outside.

# Keep the root zero

$$x^2 + 5x = x(x + 5)$$

$$x(x + 5) = 0$$

$$x = 0 \quad \text{or} \quad x = -5$$

**? Why would dividing immediately by x lose a solution?**

# An expression and an equation ask different things

| Instruction | Result |
| --- | --- |
| Factor $x^2+5x+6$ | $(x+2)(x+3)$ |
| Solve $x^2+5x+6=0$ | $x=-2$ or $x=-3$ |

An expression has equivalent forms. An equation asks which values make a statement true.

**? When can the zero-product rule identify roots?**

# Why a zero product gives two possibilities

$$AB = 0 \quad \Longrightarrow \quad A = 0 \text{ or } B = 0$$

$$(x+2)(x+3) = 0$$

For real numbers, two nonzero factors have a nonzero product. At least one factor must be zero.

**? Does AB=6 let us conclude A=0 or B=0?**

# Solve a linear equation for each factor

$$(x+2)(x+3)=0$$

$x+2=0$

$x=-2$

$x+3=0$

$x=-3$

# The bracket constant and the root have opposite signs

$$(x-2)(x-3)=0$$

$$x-2=0 \qquad x-3=0$$

$$x=2 \qquad x=3$$

**? What operation changes x−2=0 into x=2?**

CHOOSE ONE ANSWER

# Read the roots from opposite-sign factors

$$(x+3)(x-2)=0$$

$A.\quad x=3 \text{ or } -2$

$B.\quad x=-3 \text{ or } 2$

$C.\quad x=-3 \text{ or } -2$

$D.\quad x=3 \text{ or } 2$

**? Which pair solves the equation?**

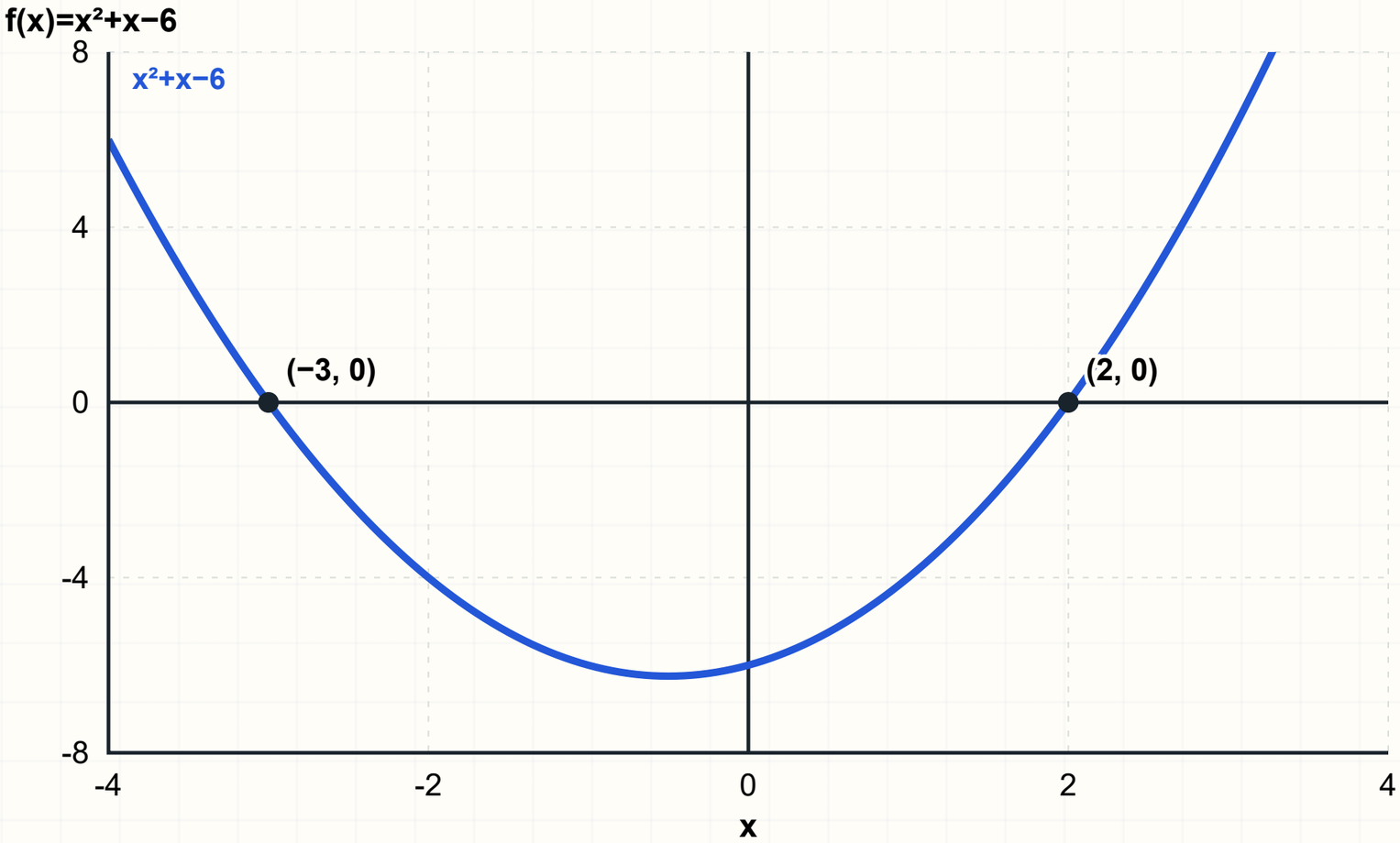
# Substitute into the original equation

$$f(x) = x^2 + x - 6$$

| Value | Substitution | Result |
|---|---|---|
| −3 | 9−3−6 | 0 |
| 2 | 4+2−6 | 0 |
| 3 | 9+3−6 | 6 |

**? Why is x=3 excluded?**

# The roots are the horizontal-axis crossings




$$(x+3)(x-2)$$

Both roots give output zero.

The graph checks the signs of the roots.

# Some quadratics have no integer factor pair

$$x^2 + x - 1$$

| Integer pair | Product | Sum |
|---|---|---|
| 1 and −1 | −1 | 0 |

No integer pair has product −1 and sum 1. Another method is needed to find its real roots.

**? Does the failed integer search prove there are no real roots?**

# Keep the same checking routine

| Step | Check |
| --- | --- |
| 1. Read | The $x^2$ coefficient is 1. Identify b and c, including their signs. |
| 2. Search | Choose signs and list integer factor pairs of c. |
| 3. Compare | Find a pair whose sum is b. For c=0, take out x. |
| 4. Verify | Expand the brackets and compare all three coefficients. |
| 5. Solve, if asked | Set each factor to zero, then check in the original equation. |

CHOOSE ONE ANSWER

# Exit check: factor before solving

$$x^2 - 7x + 12 = 0$$

$A.$ $x = -3$ or $-4$

$B.$ $x = 2$ or $6$

$C.$ $x = 3$ or $4$

$D.$ $x = -2$ or $-6$

**? Which roots survive factorisation and substitution?**

# Choose a practice rank and keep checking

| Rank | Questions | Focus |
|---|---|---|
| 1 · Foundation | 1–8 | Products, sums, signs and expansion |
| 2 · Core | 9–16 | Factor monic quadratics |
| 3 · Application | 17–22 | Roots, graphs and error diagnosis |
| 4 · Challenge (optional) | 23–24 | Missing coefficients and integer-factor limits |

Work at your own pace. Choose one answer per question. Return to the example you need.

**? Which two conditions must your chosen numbers meet?**